Unit 4 Progress Check: MCQ Part C

1. Consider the following code segment.

```
int[][] table = {{9, 8, 7}, {6, 5, 4}, {3, 2, 1}};
System.out.println(table[2][1]);
```

What value is printed as a result of executing the code segment?

(A)
2

(B)
3

(C)
4

(D)
6

2. Consider the following code segment, which is intended to create and initialize the two-dimensional array num so that columns with an even index will contain only even integers and columns with an odd index will contain only odd integers.

```
int[][] num = /* missing code */;
```

Which of the following initializer lists could replace /* missing code */ so that the code segment will work as intended?

(A)
{{0, 1, 2}, {4, 5, 6}, {8, 3, 6}}

(B)
{{1, 2, 3}, {3, 4, 5}, {5, 6, 7}}

(C)
{{2, 1, 4}, {5, 2, 3}, {2, 7, 1}}

(D)
{{2, 4, 6}, {1, 3, 5}, {6, 4, 2}}

3. The following method is intended to return the total number of elements in the 2D array arr. The method does not work as intended.

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```
public int totalElements(int[][] arr)
{
  return arr.length * arr.length;   // Line 3
}
```

Which of the following changes can be made so that the code segment works as intended?

(A) Changing line 3 to return (arr.length - 1) * (arr.length - 1);
(B) Changing line 3 to return arr.length + arr.length;
(C) Changing line 3 to return (arr.length - 1) * (arr[0].length - 1);
(D) Changing line 3 to arr.length * arr[0].length;

4. Consider the following code segment.

```
int[][] mat = {{1, 2, 3, 4},
               {5, 6, 7, 8},
               {8, 7, 6, 5},
               {4, 3, 2, 1}};

for (int r = 0; r < mat.length; r++)
{
  /* missing code */
}
```

This code segment is intended to produce the following output.

```
1
5 6
8 7 6
4 3 2 1
```

Which of the following can be used to replace `/* missing code */` so that the code segment works as intended?

(A)

```
for (int c = 0; c <= r; c++)
{
  System.out.print(mat[c][r] + " ");
}
System.out.println();
```

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(B)

```
for (int c = 0; c <= r; c++)
{
  System.out.print(mat[r][c] + " ");
}
System.out.println();
```

(C)

```
for (int c = 1; c < r; c++)
{
  System.out.print(mat[r][c] + " ");
}
System.out.println();
```

(D)

```
for (int c = 0; c <= r; c++)
{
  System.out.print(mat[r][c] + " ");
  System.out.println();
}
```

5. Consider the following code segment.

```
int[][] board = new int[5][4];

/* code to initialize board */

int total = 0;
for (int r = 1; r < board.length; r += 2)
{
  for (int c = 0; c < board[0].length; c++)
  {
    total += board[r][c];
  }
}
```

Which of the following best describes the value of total as a result of executing the code segment?

(A) It contains the sum of elements in even-numbered rows.
(B) It contains the sum of elements in odd-numbered rows.
(C) It contains the sum of elements in even-numbered columns.
(D) It contains the sum of elements in odd-numbered columns.

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6. Consider the following method, sumRows, which is intended to traverse all the rows in the two-dimensional (2D) integer array num and print the sum of all the elements in each row.

```
public static void sumRows(int[][] num)
{
  for (int[] r : num)
  {
    int sum = 0;
    for (int j = 0; j < num.length; j++)
    {
      sum += r[j];
    }
    System.out.print(sum + " ");
  }
}
```

For example, if num contains {{3, 5}, {6, 8}}, then sumRows(num) should print the following.

```
8 14
```

The method does not always work as intended.

For which of the following two-dimensional array input values does sumRows **not** work as intended?

(A)
{{10, -18}, {48, 17}}

(B)
{{-5, 2, 0}, {4, 11, 0}}

(C)
{{4, 1, 7}, {-10, -11, -12}}

(D)
{{1, 2, 3}, {4, 5, 6}, {7, 8, 9}}

7. The following method is intended to return `true` if all elements in row `x` of the 2D array `nums` are positive and `false` otherwise.

```
/**
 * Precondition: x is a valid row index of nums.
 */
public static boolean allPositiveInRow(int[][]nums, int x)
{
  boolean allPositive = /* initial value */;
  for (int y = 0; y < nums[0].length; y++)
```

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```
  {
    if (nums[x][y] <= 0)
    {
      allPositive = /* updated value */;
    }
  }
  return allPositive;
}
```

Which of the following replacements for `/* initial value */` and `/* updated value */` can be used so that the method works as intended?

(A) Replace `/* initial value */` with `true` and replace `/* updated value */` with `true`.
(B) Replace `/* initial value */` with `true` and replace `/* updated value */` with `false`.
(C) Replace `/* initial value */` with `false` and replace `/* updated value */` with `true`.
(D) Replace `/* initial value */` with `false` and replace `/* updated value */` with `false`.

8. Consider the following code segment.

```
int[][] mat = {{1, 2, 3, 4},
               {1, 2, 3, 4},
               {5, 6, 7, 8},
               {5, 6, 7, 8}};

int numRows = mat.length;
int numCols = mat[0].length;
int [][] changedMat = new int[numRows][numCols];
for (int row = 0; row < numRows; row++)
{
  for (int col = 0; col < numCols; col++)
  {
    changedMat[numRows – row – 1][col] = mat[row][col];
  }
}
```

Which of the following represents the contents of changedMat after executing this code segment?

(A)

```
{{5, 6, 7, 8},
 {5, 6, 7, 8},
 {1, 2, 3, 4},
 {1, 2, 3, 4}}
```

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(B)

{{1, 2, 3, 4},
{5, 6, 7, 8},
{1, 2, 3, 4},
{5, 6, 7, 8}}

(C)

{{4, 3, 2, 1},
{4, 3, 2, 1},
{8, 7, 6, 5},
{8, 7, 6, 5}}

(D)

{{1, 2, 3, 4},
{1, 2, 3, 4},
{5, 6, 7, 8},
{5, 6, 7, 8}}

9. In the following code segment, `board` is a properly declared and initialized 2D array of `int` values.

```
int sum = 0;
for (int r = 0; r < board.length; r++)       // Line 2
{
  for (int c = 0; c < board[0].length; c++)  // Line 4
  {
    if (c % 2 != 0)                          // Line 6
    {
      sum += board[r][c];                    // Line 8
    }
  }
}
```

The code segment is intended to compute the sum of all elements in all rows that have odd-numbered row indices. However, the code segment is not working as intended. Which of the following changes can be made so that the code segment works as intended?

(A) Changing line 2 to `for (int r = 0; r < board.length; r += 2)`
(B) Changing line 4 to `for (int c = 1; c < board[0].length; c += 2)`
(C) Changing line 6 to `if (r % 2 != 0)`
(D) Changing line 8 to `sum += board[c][r];`

10. Consider the following method.

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```
public static int findTarget(int[] nums, int target)
{
  int answer = -1;
  for (int j = nums.length - 1; j >= nums.length / 2; j--)
  {
    if (nums[j] == target)
    {
      answer = j;
    }
  }
  return answer;
}
```

The following code segment appears in another method in the same class as findTarget.

```
int[] arr = {4, 5, 8, 3, 8, 9, 8, 1};
System.out.println(findTarget(arr, 8));
```

What is printed as a result of executing the code segment?

(A) 2
(B) 4
(C) 5
(D) 6

11. Consider the following method.

```
public static int someSearch(int[] arr, int value)
{
  for (int j = 0; j < arr.length; j++)
  {
    if (arr[j] == value)
    {
      return j;
    }
  }
  return -1;
}
```

The following code segment appears in a method in the same class as `someSearch`.

```
int[] myNumbers = {1, 2, 5, 2, 6, 3, 3, 5, 3};
System.out.println(someSearch(myNumbers, 3));
```

What is printed as a result of executing the code segment?

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(A)
-1

(B)
5

(C)
6

(D)
8

12. Consider the method `linearSearch`, which takes an `ArrayList` of `Integer` elements and a target `int` value as parameters and returns the index of the first appearance of the target value in the list or `-1` if the target value does not appear in the list.

```
public static int linearSearch(ArrayList<Integer> elements, int target)
{
  for (int j = 0; j < elements.size(); j++)    // Line 3
  {
    if (elements.get(j) == target)
    {
      return j;
    }
  }
  return -1;
}
```

Which of the following describes how replacing line 3 with `for (int j = (elements.size() - 1); j >= 0; j--)` will affect the behavior of `linearSearch`?

(A) The modification has no effect: the modified method will continue to return the index of the first appearance of the target value in the list, or `-1` if the target value does not appear in the list.

(B) The modified method will return the index of the last appearance of the target value in the list, or `-1` if the target value does not appear in the list.

(C) The modified method will throw an `IndexOutOfBoundsException`.

(D) The modified method will return `-1` regardless of the inputs.

13. The following method is a correct implementation of the selection sort algorithm. The method correctly sorts the elements of `arr` so that they appear in order from least to greatest.

```
public static void selectionSort(int[] arr)
{
  for (int j = 0; j < arr.length - 1; j++)
  {
    int minIndex = j;
    for (int k = j + 1; k < arr.length; k++)
    {
```

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```
      if (arr[k] < arr[minIndex])
      {
        minIndex = k;
      }
    }
    int temp = arr[j];
    arr[j] = arr[minIndex];
    arr[minIndex] = temp;
    /* end of outer loop */
  }
}
```

Which of the following could be the contents of arr after two passes of the outer loop (i.e., when $j == 1$ at the point indicated by /* end of outer loop */)?

(A)

{10, 20, 50, 40, 30}

(B)

{10, 30, 50, 40, 20}

(C)

{20, 10, 50, 40, 30}

(D)

{20, 30, 50, 40, 10}

14. The following method is intended to sort an array of integers into ascending order.

```
public static void sort(int[] arr)
{
  for (int j = 0; j < arr.length - 1; j++)
  {
    int minIndex = j;
    for (int k = j + 1; k < arr.length - 1; k++)
    {
      if (arr[k] < arr[minIndex])
      {
        minIndex = k;
      }
    }
    int temp = arr[minIndex];
    arr[minIndex] = arr[j];
    arr[j] = temp;
  }
}
```

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This method works as intended for some, but not all, int arrays. Which of the following arrays can be used to show that the method does not work as intended?

(A)
{1, 2, 3, 4, 5}

(B)
{1, 2, 3, 5, 4}

(C)
{1, 2, 4, 3, 5}

(D)
{1, 3, 2, 4, 5}

15. Consider the following correct implementation of the selection sort algorithm.

```
public static void selectionSort(int[] elements)
{
  for (int j = 0; j < elements.length - 1; j++)
  {
    int minIndex = j;
    for (int k = j + 1; k < elements.length; k++)
    {
      if (elements[k] < elements[minIndex])
      {
        minIndex = k;
      }
    }
    if (j != minIndex)
    {
      int temp = elements[j];
      elements[j] = elements[minIndex];
      elements[minIndex] = temp;   // line 17
    }
  }
}
```

The following declaration and method call appear in a method in the same class as selectionSort.

```
int[] arr = {30, 40, 10, 50, 20};
selectionSort(arr);
```

How many times is the statement elements[minIndex] = temp; in line 17 of the method executed as a result of the call to selectionSort?

(A) 2

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(B) 3
(C) 4
(D) 5

16. Consider the following method.

```
public static int mystery(int n)
{
  if (n < 10)
  {
    return n;
  }
  return (n % 10) + mystery(n / 10);
}
```

Assume that the `int` variable `num` is properly declared and initialized to a value greater than 0. Which of the following best describes the value returned by the call `mystery(num)`?

(A) The value of `num`
(B) The rightmost digit of `num`
(C) The sum of the digits of `num`
(D) Two times the rightmost digit of `num`

17. The following recursive method is intended to return `true` if `origNum` is divisible by each digit in `numsToDivideBy`, and to return `false` otherwise. For example, `isDivisible(236, 24)` should return `true` because `236` is divisible by both `2` and `4`. The method does not always work as intended.

```
public static boolean isDivisible(int origNum, int numsToDivideBy)
{
  if (numsToDivideBy == 0)
  {
    return true;
  }
  else if (origNum % (numsToDivideBy % 10) != 0)
  {
    return false;
  }
  else
  {
    return isDivisible(origNum, numsToDivideBy / 10);
  }
}
```

Which of the following method calls can be used to demonstrate that the method does **not** work as intended?

(A)
isDivisible(120, 12)

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(B)
isDivisible(135, 5)

(C)
isDivisible(225, 105)

(D)
isDivisible(296, 341)

18. Consider the following recursive method.

```
public static int mystery(int x)
{
  if (x <= 0)
  {
    return 5;
  }
  else
  {
    return x + mystery(x - 2);
  }
}
```

What is returned as a result of the method call `mystery(5)`?

(A)
8

(B)
13

(C)
14

(D)
20

19. The following recursive method is intended to return the index of the first occurrence of goal in arr between index start and index end, inclusive. The index of the first occurrence of goal is the index closest to start where goal is found. If goal does not appear in the range of indices from start to end, the method returns -1. However, the method does not work as intended.

```
/** Precondition: 0 <= start <= end, end < arr.length */
public static int locateItem(int[] arr, int goal, int start, int end)
{
```

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```
  if (start == end)                          // Line 4
  {
    return -1;
  }
  else if (arr[start] == goal)               // Line 8
  {
    return start;                            // Line 10
  }
  else
  {
    return locateItem(arr, goal, start + 1, end);  // Line 14
  }
}
```

Which of the following changes can be made so that the method works as intended?

(A) Changing line 4 to `if (start > end)`
(B) Changing line 8 to `else if (start == goal)`
(C) Changing line 10 to `return arr[start];`
(D) Changing line 14 to `return locateItem(arr, goal, start + 1, end + 1);`

20. Consider the following method.

```
public static void find(String[] words, int s, int i)
{
  if (i != words.length)
  {
    if (i == 0 || !(words[s].equals(words[i])))
    {
      s = i;
      System.out.print(s + " ");
    }
    find(words, s, i + 1);
  }
}
```

The following code segment appears in a method in the same class as find.

```
String[] furniture = /* missing initialization */ ;
find(furniture, 0, 0);
```

Which of the following can be used to replace /* missing initialization */ so that the code segment prints 0 1 4 6 7 ?

(A)

```
{"Stool", "Chair", "Chair", "Chair",
 "Stool", "Couch", "Couch", "Table"}
```

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(B)
{"Table", "Couch", "Couch", "Stool",
"Chair", "Chair", "Chair", "Stool"}

(C)
{"Stool", "Chair", "Chair", "Chair",
"Couch", "Couch", "Stool", "Table"}

(D)
{"Table", "Stool", "Couch", "Couch",
"Chair", "Chair", "Chair", "Stool"}

21. Consider the following recursive method.

```
public static String doSomething(String str)
{
  if (str.length() < 1)
  {
    return "";
  }
  else
  {
    return str.substring(0, 1) + doSomething(str.substring(1));
  }
}
```

Assume that `myString` is a properly declared and initialized `String` object. Which of the following best describes the result of the call `doSomething(myString)`?

(A) The method call returns a `String` containing the contents of `myString` unchanged.
(B) The method call returns a `String` containing the contents of `myString` with the order of the characters reversed from their order in `myString`.
(C) The method call returns a `String` containing all but the first character of `myString`.
(D) The method call returns a `String` containing only the first and second characters of `myString`.